

SUSHANT GARG

Forward-Deployed AI / Machine Learning & Generative AI

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SKILLS

Large Language Models: LLM Application Design, Retrieval-Augmented Generation (RAG), Prompt Engineering, Few-shot & Chain-of-Thought, In-context Learning, Embeddings & Vector Databases (FAISS, Pinecone, Chroma), Semantic Search, Fine-tuning, LoRA / PEFT, Instruction Tuning, RLHF, Quantization, Context-window Optimization

LLM Systems & Agents: Agentic Workflows, Multi-Agent Orchestration, Function Calling & Tool Use, Model Context Protocol (MCP), LLM Evaluation (faithfulness, relevance, hallucination), Guardrails & Safety, LLM Observability, Prompt & Token Cost Optimization, LLMops

LLM Tooling: OpenAI, Anthropic Claude, Hugging Face Transformers, LangChain, LlamaIndex, LangGraph, Ollama, vLLM, Weights & Biases

Deep Learning & NLP: Neural Networks, Transformers, Attention, Named Entity Recognition, Text Classification, Topic Modeling, Sentiment Analysis, Summarization, Sequence Models

Machine Learning: Regression, Classification, Clustering, Time-Series Forecasting (SARIMAX), Uplift & Propensity Modeling, Reinforcement Learning, Multi-Armed Bandits, Recommendation, Optimization

Statistics & Experimentation: Causal Inference, A/B Testing, CUPED, Bayesian Inference, Synthetic Controls, Elasticity Modeling, Hypothesis Testing

MLOps & Data Engineering: Databricks, Spark, Airflow, MLflow, Feature Stores, Model Monitoring, Challenger Models, CI/CD for ML, Kafka, Snowflake

Languages & Frameworks: Python, R, SQL, PySpark, PyTorch, scikit-learn, pandas, NumPy, LangChain

Cloud & BI: AWS, Azure, GCP, Tableau, Power BI

Leadership: Team Leadership, AI Strategy & Roadmap, Stakeholder Management, Executive Communication, Mentoring, Agile Execution

EDUCATION

B.Tech, Computer Science & Engineering

SRM University

Chennai, India

EXPERIENCE

Associate Manager, Forward-Deployed Generative AI & Data Science

Majid Al Futtaim, Carrefour

Aug 2022 – Present

Dubai, UAE | Remote

- **GenAI Team Leadership:** Led a team of 4–5 junior data scientists building LLM applications on Databricks, owning the GenAI roadmap, evaluation standards, and LLMops practices, raising delivery speed by 30%.
- **Forward-deployed Partnership:** Embedded directly with country teams, commercial, category management, account management, and operations to translate business problems into GenAI/ML solutions and drive adoption.
- **RAG Analytics Assistant:** Built a retrieval-augmented generation assistant over pricing, promotion, and sales data using embeddings and vector search, enabling natural-language querying and self-serve insights that cut manual reporting effort.
- **Conversational Insight & Reporting:** Developed LLM pipelines for automated narrative generation, executive summaries, and report drafting, combining prompt engineering with guardrails for factual accuracy.
- **Agentic Workflows:** Designed multi-agent and function-calling systems that orchestrate analytics tools to answer complex commercial questions end to end.
- **LLM Evaluation & Fine-tuning:** Built evaluation harnesses scoring faithfulness, relevance, and hallucination, and applied fine-tuning and LoRA to domain-adapt models to retail terminology.
- **Semantic Search & Embeddings:** Stood up a vector database of products and documents to power semantic product discovery and similarity matching.
- **GenAI-assisted Optimization:** Paired LLM reasoning with elasticity and Bayesian models to recommend promo strategies, driving \$8M annual savings and 20% higher promo ROI.
- **Text-to-SQL & Self-serve BI:** Built an LLM text-to-SQL layer over the analytics warehouse, letting business users query data in plain English and reducing ad-hoc requests to the data team.

- **Document Intelligence:** Developed **LLM** pipelines to extract and structure terms from supplier contracts and promotion agreements, cutting manual review time.
- **Prompt & Cost Optimization:** Tuned prompts, caching, and model routing to balance quality and cost across GenAI workloads, lowering token spend while holding accuracy.
- **Forecasting & Inventory ML:** Built demand-forecasting, replenishment, and stockout-prediction models with monitoring and challenger evaluation, reducing stockouts by **24%**.
- **Customer Value Modeling:** Built segmentation, cohort, and propensity models on large loyalty and transaction datasets to power personalized targeting.

Data Scientist 2 (NLP & Machine Learning)

Mar 2021 – Aug 2022

SWVL

Dubai, UAE

- **NLP for Customer Feedback:** Built **text classification** and **sentiment analysis** models on rider reviews and support tickets to surface churn drivers, informing retention strategy that lowered churn by **25%**.
- **Support Ticket Intelligence:** Developed **NER** and **topic-modeling** pipelines to auto-categorize and route support conversations, reducing manual triage.
- **Conversational Analytics:** Built **transformer-based** embedding and **semantic-search** models over rider chat logs to cluster intents and detect emerging issues.
- **Text Summarization:** Built abstractive summarization of long support threads to give operations teams faster context on recurring problems.
- **Pricing & Marketplace ML:** Built **multi-armed bandit** and adaptive-experimentation pricing models, lifting revenue by **12%** and marketplace efficiency by **18%**.
- **NLP Pipelines & MLOps:** Built reusable text-preprocessing, embedding, and model-serving pipelines, cutting deployment time by **50%** and failures by **35%**.

Data Analyst (ML & Analytics)

Jan 2020 – Mar 2021

Blinkit (Grofers)

Bengaluru, India

- **Time-Series Forecasting:** Built **SARIMAX**, ML, and demand-sensing forecasting models with vintage and trend diagnostics, cutting inventory holding by **60%** and lifting availability from **55% to 85%**.
- **Predictive Replenishment:** Built predictive, exception-based replenishment and SKU-level demand models that reduced out-of-stock incidents by **18%**.
- **Pricing ML:** Built price-elasticity and promotion-lift models for hyperlocal pricing decisions, driving **12% GMV** growth.
- **Clustering & Assortment:** Built clustering and seasonality-based category models to lift SKU productivity, delivering **15% category growth**.
- **Simulation & Optimization:** Built warehouse simulation and operational-optimization models, improving pick-ing speed by **25%** and saving **20%** on labor.

Decision Scientist (ML & Data Engineering)

Jul 2018 – Jan 2020

Mu Sigma

Bengaluru, India

- **Predictive Modeling:** Built regression, classification, and time-series models for Fortune 100 telecom clients on acquisition, retention, and digital adoption, raising digital adoption by **10%** and model accuracy by **20%**.
- **NLP & Text Analytics:** Applied NLP and text-mining techniques to customer-interaction data to extract drivers of churn and digital adoption.
- **ML Data Pipelines:** Built real-time **Kafka-to-Snowflake** streaming pipelines feeding ML models for **40M+** customers under strict data governance.
- **Automation:** Automated schema generation and reusable ML assets, saving 20 to 300 hours per implementation and speeding execution by **30%**.

PERSONAL PROJECTS

Triax : Retail & CPG Revenue Margin Intelligence Platform

2025

<http://35.209.74.91> (*live*)

Solo Build

- Built a margin-intelligence platform that goes beyond volume-based RGM to optimize a **“3-C” constraint**, balancing Company margin, Customer (retailer) margin, and Consumer willingness-to-pay on every pricing, promotion, and assortment decision.
- Shipped **44 features across 24 modules**, including price elasticity, trade-promotion ROI, **agent-based shopper simulation**, gross-to-net controls, and guardrails.
- Built an **RGM Copilot AI agent** that answers plain-language questions by **function-calling** the right analytics engines and returning grounded results.
- Engineered with **FastAPI**, **React/TypeScript**, **PostgreSQL**, and **scikit-learn**, fully containerized